POWER BI PROJECT | DATA ENGINEERING BOOTCAMP

Retail Sales Analytics Dashboard

Sales, products, customers and countries in one interactive five-page Power BI report

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90.58M	26.63M	2,000	80
Total sales	Total profit (29% margin)	Customers	Products in 4 categories

Tools: Excel | Power Query (M) | DAX | Power BI Desktop

Data: 8 tables, about 49,500 invoices, 4 countries, 12 cities

1. Project at a glance

This project turns eight raw sales tables into an interactive Power BI report for a retail store that sells electronics, home appliances, clothing and books in four countries. The report is built from one home page and four analysis pages, and every page answers a clear business question.

Page	Question it answers	Main visuals
Executive Overview	How are sales and profit performing?	KPI cards, sales and profit trend, sales by category, top products
Product Analysis	What makes money and what loses it?	Sales by sub-category, volume vs margin scatter, top products by profit, product table
Customer Analysis	Who buys, and who comes back?	New vs active customers, sales by gender, age groups, top 10 customers
Country & Location	Where are we selling?	City map, sales by country, cities ranked, country by category heatmap

How the project was built

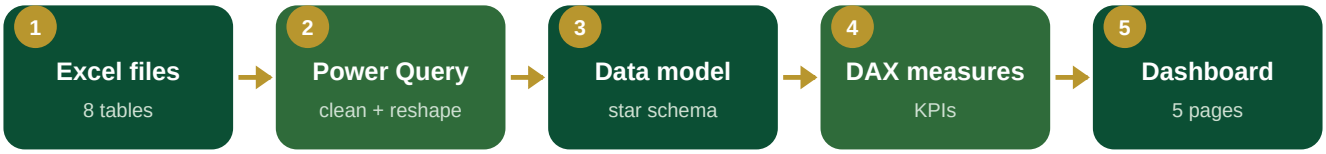


Figure 1. From raw Excel files to an interactive dashboard

The five steps: the Excel files were loaded into Power Query, where the data was cleaned and reshaped; the tables were connected in a star-schema data model; DAX measures calculated every KPI; and the measures were used to design five report pages with slicers and navigation.

2. The data and how I prepared it

2.1 Source tables

Table	Rows	What it holds
Fact_Sales	about 49.5K	One row per invoice line: invoice, date, customer, product, quantity, unit price, total amount
Dim_Customer	2,000	Name, gender, age, registration date and location of each customer
Dim_Product	80	Product name, sub-category, price and cost
Dim_SubCategory / Dim_Category	9 / 4	Electronics, Home Appliances, Clothing, Books and their sub-categories
Dim_Location / Dim_Country	12 / 4	12 cities in Egypt, Saudi Arabia, USA and Germany, with a region for each country
Dim_Date	1,000 days	Original calendar table (replaced during preparation, see below)

2.2 Data quality log

Each table was profiled in Power Query (valid, error and empty percentages, distinct and unique counts). These are the problems found and how each was handled.

Issue found	Impact	Action taken
Duplicate invoice rows	Sales would be counted twice	Removed duplicates using Invoice + Customer + Product + Quantity as the key
500 rows (about 1%, 0.93M in sales) with no CustomerID and no ProductID	They cannot be linked to a customer or product and showed up as a (Blank) category	Filtered out of Fact_Sales and documented
Date table too short: 1,000 days from January 2021 ends in 2023, but invoices run to 2025	Later sales would have no date; trends and year-over-year would be wrong	Built a new calendar in Power Query from the first to the last year of sales
Repeated names: 80 products share 31 names; 2,000 customers share 128 names	Charts would merge different products or customers	Created Product Label and Customer Label (name + ID)
Age missing for 5% of customers	These customers could disappear from age charts	Kept the rows and grouped them as "Unknown"
Category and location split across several tables (snowflake)	Longer filter chains and a model that is not a pure star	Merged category data into Dim_Product and location data into Dim_Customer

2.3 Power Query steps (M)

- **Merge queries** (left outer join) to flatten sub-category, category, city and country into the dimensions.
- **Custom columns** for Product Label, Customer Label, Age Group and Age Group Sort (so age groups appear in age order).
- **New calendar table** generated with List.Dates from the first to the last year in the sales data, with Year, Quarter, Month and Year-Month columns.
- **Data types and staging:** correct types for every column, and raw tables kept as staging queries (named src_) so only clean tables are loaded.

3. The data model

The model is a **star schema**: one fact table in the center and three dimension tables around it. Every dimension connects directly to the fact table, so filters always flow in one simple direction: from the dimension to the sales.

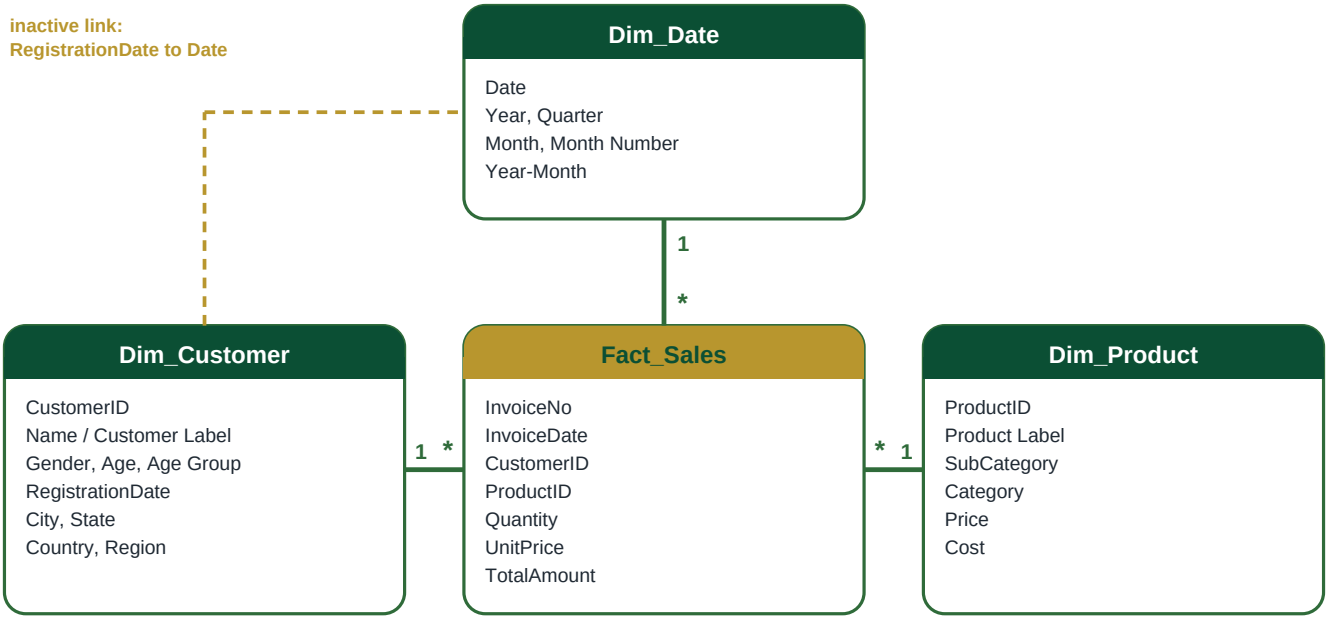


Figure 2. Star schema of the report (solid lines: active relationships, dashed line: inactive)

From (many side)	To (one side)	Type
Fact_Sales[CustomerID]	Dim_Customer[CustomerID]	Active, one-to-many
Fact_Sales[ProductID]	Dim_Product[ProductID]	Active, one-to-many
Fact_Sales[InvoiceDate]	Dim_Date[Date]	Active, one-to-many
Dim_Customer[RegistrationDate]	Dim_Date[Date]	Inactive, used only by the New Customers measure

Why a star schema?

- **Simple filters:** a slicer on Category or Country filters the sales directly, with no long chain of tables.
- **Fast and easy to read:** fewer tables and relationships make the model easier to understand and maintain.
- **Role-playing date:** the date table is used twice, once for the invoice date (active) and once for the customer registration date (inactive, switched on by a DAX function when needed).

Assumptions
"Country" means the country where the customer lives, because the data has no store or shipping location. All amounts are treated as one currency, because the data has no currency column.

4. DAX measures

All measures are stored in a dedicated **_Measures** table, so they are easy to find. A measure is calculated for the current filter selection, so the same measure answers different questions on different pages.

4.1 Sales and profit

Measure	Formula	What it answers
Total Sales	`SUM(Fact_Sales[TotalAmount])`	Revenue in the current selection
Total Quantity	`SUM(Fact_Sales[Quantity])`	Units sold
Total Orders	`DISTINCTCOUNT(Fact_Sales[InvoiceNo])`	Invoices, each counted once
Total Cost	`SUMX(Fact_Sales, Fact_Sales[Quantity] * RELATED(Dim_Product[Cost]))`	Cost of goods sold, row by row
Total Profit	`[Total Sales] - [Total Cost]`	Money left after cost
Profit Margin	`DIVIDE([Total Profit], [Total Sales])`	Profit as a share of sales
Avg Selling Price	`DIVIDE([Total Sales], [Total Quantity])`	Average price per unit

4.2 Time intelligence

Measure	Formula	What it answers
Sales LY	`CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Dim_Date[Date]))`	Sales in the same period last year
YoY %	`DIVIDE([Total Sales] - [Sales LY], [Sales LY])`	Growth compared with last year
Sales YTD	`TOTALYTD([Total Sales], Dim_Date[Date])`	Running total from 1 January

4.3 Customers

Measure	Formula	What it answers
Active Customers	`DISTINCTCOUNT(Fact_Sales[CustomerID])`	Customers who bought
Total Customers	`COUNTROWS(Dim_Customer)`	Registered customers
New Customers	`CALCULATE(COUNTROWS(Dim_Customer), USERELATIONSHIP(Dim_Customer[RegistrationDate], Dim_Date[Date]))`	Customers by registration date
Repeat Customers	`COUNTROWS(FILTER(VALUES(Fact_Sales[CustomerID]), [Total Orders] > 1))`	Customers with more than one order
Repeat Customer Rate	`DIVIDE([Repeat Customers], [Active Customers])`	Share of customers who came back
Avg Orders per Customer	`DIVIDE([Total Orders], [Active Customers])`	Purchase frequency
Sales per Customer	`DIVIDE([Total Sales], [Active Customers])`	Average spend per customer

4.4 Rankings, products and locations

Measure	Formula	What it answers
Top Product / Category / Country / City	`MAXX(TOPN(1, VALUES(Dim_Product[Product Label]), [Total Sales]), Dim_Product[Product Label])`	Name of the best performer (same pattern for each)
Top Country Share	`DIVIDE(MAXX(TOPN(1, VALUES(Dim_Customer[Country]), [Total Sales]), [Total Sales]), [Total Sales])`	Dependence on the best market

Measure	Formula	What it answers
Active Countries	`COUNTROWS(FILTER(VALUES(Dim_Customer[Country]), [Total Sales] > 0))`	Countries with sales
Lowest Margin Product	`MAXX(TOPN(1, SoldProducts, [Profit Margin], ASC), Dim_Product[Product Label])`	Weakest product by margin
Sales % of Total	`DIVIDE([Total Sales], CALCULATE([Total Sales], ALLSELECTED()))`	Share of the selected sales

4.5 Key DAX ideas used

Idea	Used in	Why
SUM vs SUMX	Total Sales vs Total Cost	SUM adds one column. SUMX goes row by row, so it can multiply Quantity by a Cost taken from another table with RELATED.
CALCULATE	Sales LY, New Customers	Changes the filter context, for example by shifting the dates one year back.
DIVIDE	Every ratio	Safe division: returns blank instead of an error when the denominator is zero.
USERELATIONSHIP	New Customers	Switches on the inactive registration-date relationship for one measure only.
TOPN + MAXX	Top Product and similar	TOPN finds the best row and MAXX returns its name as text for a KPI card.

5. Report design

5.1 Color palette

Dark green	Green	Gold	Light background	Text
#0B4F34	#2F6B3A	#B8962E	#F2F3EC	#1F2933

Dark green and gold are the brand colors: green for the navigation bar and chart bars, gold for highlights and the active page. The light background makes the white cards stand out.

5.2 Navigation and structure

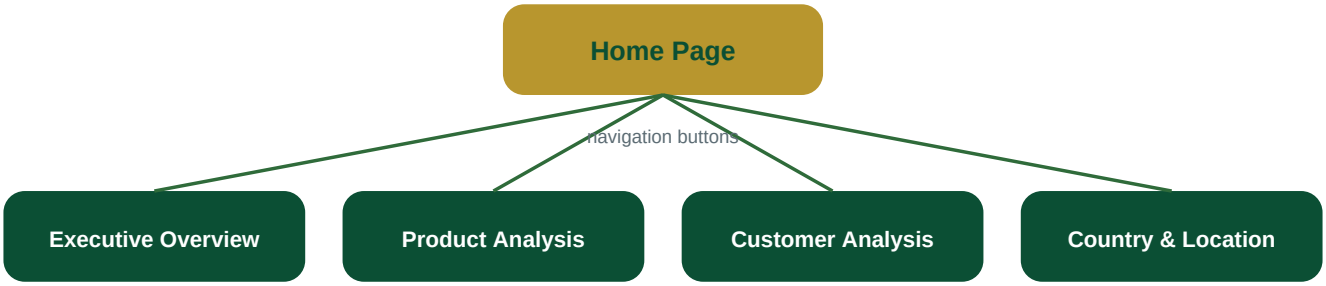


Figure 3. The home page links to the four analysis pages

5.3 Design principles

- **Same layout on every page:** a title bar with slicers on top, a row of KPI cards, then charts on rounded white cards.
- **KPI card structure:** a colored accent bar, a short title, the value, and an icon in a soft circle. Cards alternate green and gold accents.
- **Navigation bar:** a dark green bar on the left with one icon per page; the active page is highlighted in gold.
- **Slicers:** dropdown slicers for Year, Category and location, kept small so the charts get the space.
- **Interactivity:** clicking a bar, slice or bubble filters the other visuals on the page, and ranking charts use a Top N filter so they stay readable.
- **Readable numbers:** values shown in millions, percentages formatted as percentages, and sorted axes (months in date order, ages in age order).

6. The report pages

6.1 Executive Overview

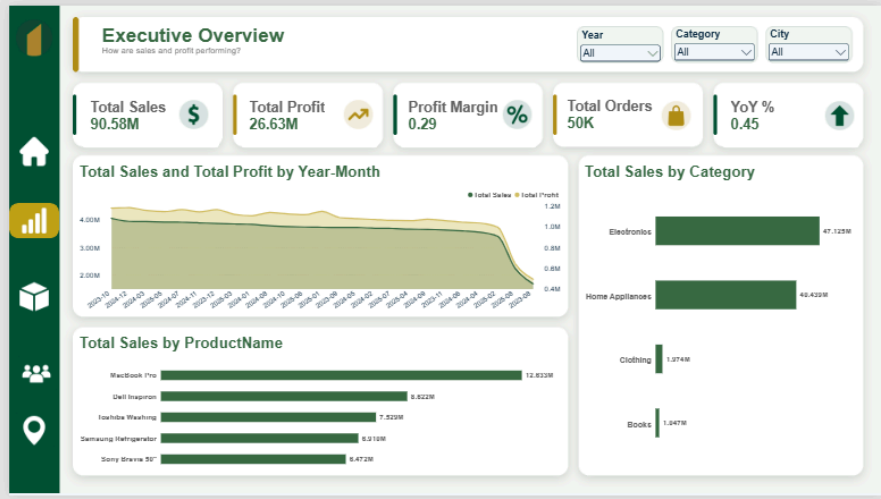


Figure 4. Executive Overview page

KPI cards	Total Sales, Total Profit, Profit Margin, Total Orders, YoY %
Visuals	Area chart of sales and profit by month; bar chart of sales by category; bar chart of top products by sales
Slicers	Year, Category, City
Takeaway	The business earns a 29% margin; two categories and a handful of premium products drive almost all revenue.

6.2 Product Analysis



Figure 5. Product Analysis page

KPI cards	Total Quantity, Avg Selling Price, Top Product, Lowest Margin
Visuals	Bar chart of sales by sub-category; scatter plot of volume against margin (bubble size = sales); top 5 products by profit; product table with data bars and a color scale on margin
Purpose	Shows which products earn money and which sell a lot but leave little profit.

6.3 Customer Analysis

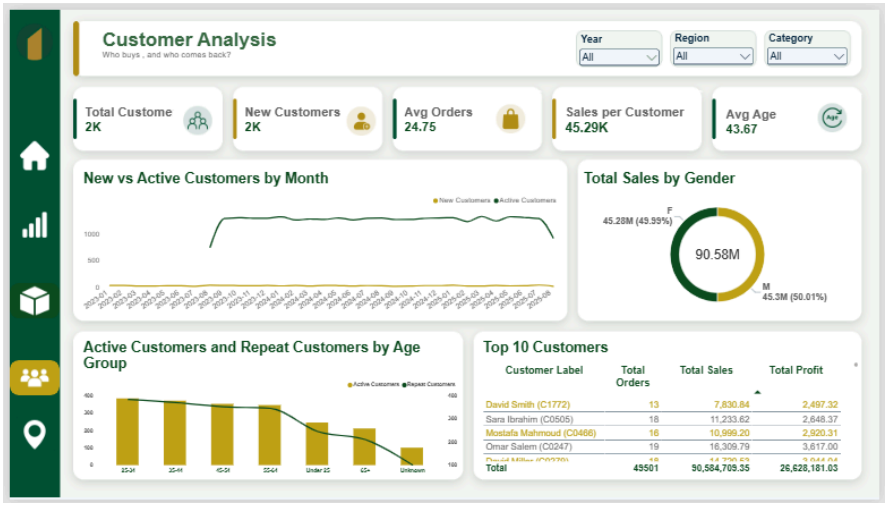


Figure 6. Customer Analysis page

KPI cards	Total Customers, New Customers, Avg Orders, Sales per Customer, Avg Age
Visuals	Line chart of new and active customers by month; donut of sales by gender; combo chart of customers by age group; top 10 customers table
Purpose	Describes who the customers are and how often they buy.

6.4 Country & Location



Figure 7. Country & Location page

KPI cards	Active Countries, Top Country, Top City, Top Country %, Profit Margin
Visuals	Bubble map of sales by city; donut of sales by country; bar chart of cities ranked by sales; matrix of country by category with a color gradient (heatmap)
Purpose	Shows where sales come from and which category sells in which market.

7. Key insights

Finding	Evidence	What it means
Two categories drive almost all revenue	Electronics 52% and Home Appliances 45% of sales; Clothing 2.2% and Books 1.2%	Clothing and Books are low-price items, so they add little revenue even when they sell in volume.
Sales are balanced across markets	Saudi Arabia 26.2%, USA 25.7%, Germany 24.7%, Egypt 23.4%	The business does not depend on one country; each market has room to grow.
Premium products lead	MacBook Pro 12.8M, Dell Inspiron 8.6M, Toshiba Washing 7.5M	A few high-value products generate a large share of revenue.
Customers buy again and again	About 25 orders per customer on average	Almost every customer is a repeat buyer, so growth should focus on value per order.
Healthy profit	26.6M profit on 90.6M sales (29% margin)	The margin is solid; the Product page shows which items pull it down.
No gender skew	Female 49.99% and male 50.01% of sales	Marketing does not need to target one gender.

8. Limitations and notes

- The data looks **synthetic** (for example, names and genders do not always match), so conclusions are for practice, not for a real business.
- The sales trend drops sharply in the last months. This probably means those months are incomplete, so they should be checked before drawing conclusions.
- Age is empty for 5% of customers. They are shown as "Unknown" and ignored by the average age.

9. What I learned

- Profiling and cleaning data in **Power Query** and documenting every decision.
- Designing a **star schema** and flattening a snowflake model.
- Writing **DAX** measures and designing a **consistent, interactive report** with a clear question for every page.